Bernoulli Equation

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Fluid Mechanics Lecture Notes

Learning Objectives

By the end of this lecture, students should be able to:

- describe a streamline and resolve acceleration into tangential and normal components;
- derive Bernoulli's equation from Newton's second law along a streamline;
- identify the assumptions required for the basic Bernoulli equation;
- interpret pressure head, velocity head, and elevation head;
- apply Bernoulli's equation together with continuity to reservoirs, pipes, nozzles, and Pitot tubes;
- recognize situations in which pumps, turbines, viscosity, or unsteady effects require an extended energy equation.

1 Why Bernoulli's Equation Matters

Bernoulli's equation connects three forms of mechanical energy in a flowing fluid:

$$\boxed{\text{pressure energy} + \text{kinetic energy} + \text{potential energy}}$$

It allows us to determine pressure, velocity, or elevation at one location when enough information is known at another location along the same streamline.

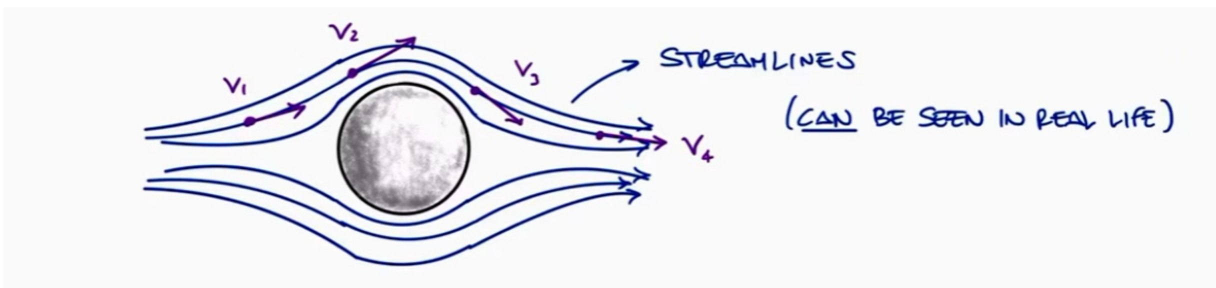
Typical applications include:

- flow from a tank or reservoir;
- nozzles and diffusers;
- Venturi meters and orifice meters;
- Pitot tubes and airspeed measurements;

- siphons and pipe systems;
- flow around vehicles, wings, and other bodies.

2 Streamlines and Flow Acceleration

A **streamline** is a curve that is tangent to the local velocity vector everywhere in the flow field. For steady flow, fluid particles follow fixed streamlines.



The acceleration of a fluid particle can be resolved into tangential and normal components:

$$\boldsymbol{a} = a_s \boldsymbol{e}_s + a_n \boldsymbol{e}_n$$

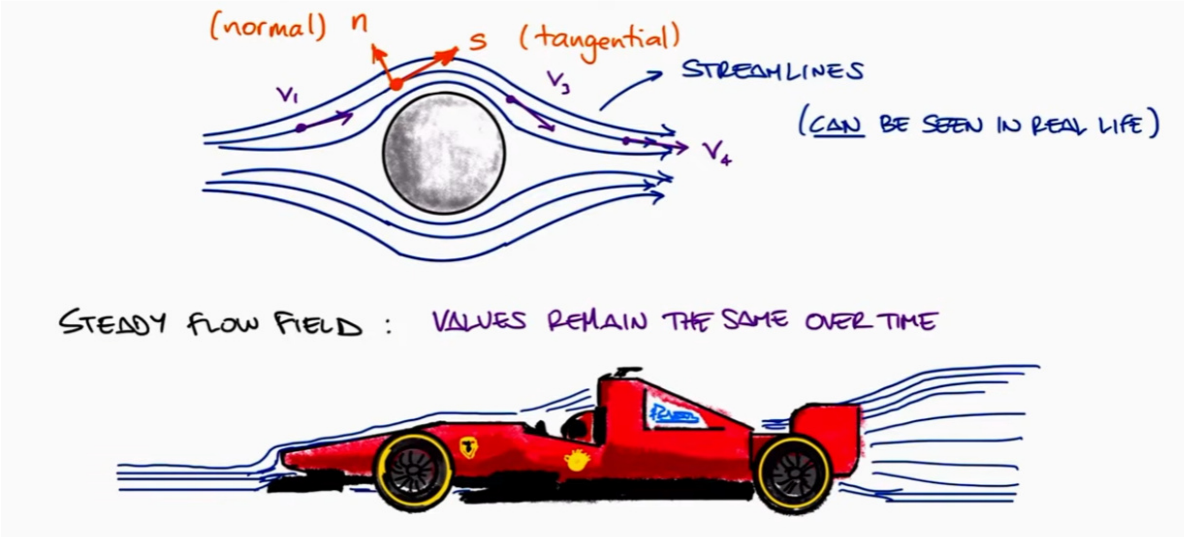
For steady flow,

$$a_s = \frac{dV}{dt} = \frac{dV}{ds}\frac{ds}{dt} = V\frac{dV}{ds}$$

and the normal acceleration is the familiar centripetal acceleration,

$$a_n = \frac{V^2}{R}$$

where R is the local radius of curvature of the streamline.



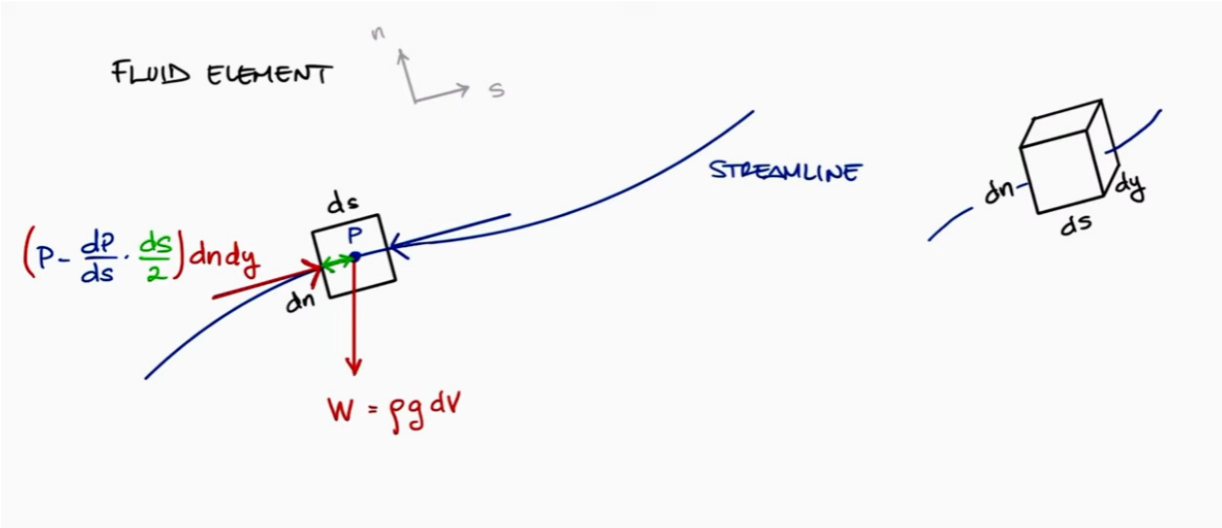
Physical Interpretation

- a_s changes the *magnitude* of velocity.
- a_n changes the *direction* of velocity.
- Bernoulli's basic derivation uses the force balance in the s-direction.

3 Derivation Along a Streamline

Consider an infinitesimal fluid element with length ds, normal dimension dn, and width dy. Its volume is

$$d\mathcal{V} = ds\,dn\,dy.$$



The pressure force resultant in the streamline direction is

$$-\frac{\partial P}{\partial s}\, d\mathcal{V}.$$

The component of the weight force in the s-direction is

$$-\rho g \sin\theta \, d\mathcal{V}.$$

Since

$$\sin\theta = \frac{dz}{ds},$$

Newton's second law along the streamline gives

$$-\frac{\partial P}{\partial s} d\mathcal{V} - \rho g \frac{dz}{ds} d\mathcal{V} = \rho d\mathcal{V} \left(V \frac{dV}{ds} \right).$$

Canceling $d\mathcal{V}$ and rearranging,

$$\frac{dP}{ds} + \rho V \frac{dV}{ds} + \rho g \frac{dz}{ds} = 0.$$

Multiplying by ds,

$$dP + \rho V\, dV + \rho g\, dz = 0.$$

For an incompressible fluid, ρ is constant. Integrating gives

$$\int dP + \rho \int V\, dV + \rho g \int dz = 0,$$

so

$$\boxed{P + \frac{1}{2}\rho V^2 + \rho g z = C}$$

along a streamline.

Between two points on the same streamline,

$$\boxed{P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2}$$

4 Assumptions

The basic Bernoulli equation used in this lecture assumes:

1. **Steady flow:** flow properties at a fixed point do not change with time.
2. **Incompressible fluid:** density is constant.
3. **Inviscid behavior:** viscous shear and friction losses are negligible.
4. **Along the same streamline:** points 1 and 2 lie on one streamline.
5. **No shaft work between the points:** no pump or turbine adds or removes energy.
6. **Gravity is the only important body force.**

For an **irrotational** flow, the Bernoulli constant is the same throughout the entire connected flow field, not only along one streamline.

When Bernoulli Cannot Be Used Directly

Do not apply the basic Bernoulli equation blindly across:

- a pump, fan, compressor, or turbine;
- a long pipe with significant friction;
- a hydraulic jump, mixing region, or strongly separated wake;
- a rapidly changing unsteady flow;
- a compressible gas flow with large density changes.

Use the extended mechanical-energy equation when these effects are important.

5 Three Common Forms

5.1 Pressure Form

$$\boxed{P + \frac{1}{2}\rho V^2 + \rho g z = \text{constant}}$$

Each term has units of pressure:

$$[P] = \text{Pa}, \qquad \left[\frac{1}{2}\rho V^2\right] = \text{Pa}, \qquad [\rho g z] = \text{Pa}.$$

Term	Meaning
P	static pressure
$\frac{1}{2}\rho V^2$	dynamic pressure
$\rho g z$	elevation or hydrostatic contribution
$P + \frac{1}{2}\rho V^2$	stagnation pressure when elevation is unchanged

5.2 Head Form

Dividing by ρg,

$$\boxed{\frac{P}{\rho g} + \frac{V^2}{2g} + z = \text{constant}}$$

Each term has units of length:

- $\dfrac{P}{\rho g}$: pressure head;
- $\dfrac{V^2}{2g}$: velocity head;
- z: elevation head.

5.3 Energy per Unit Mass Form

Dividing by ρ,

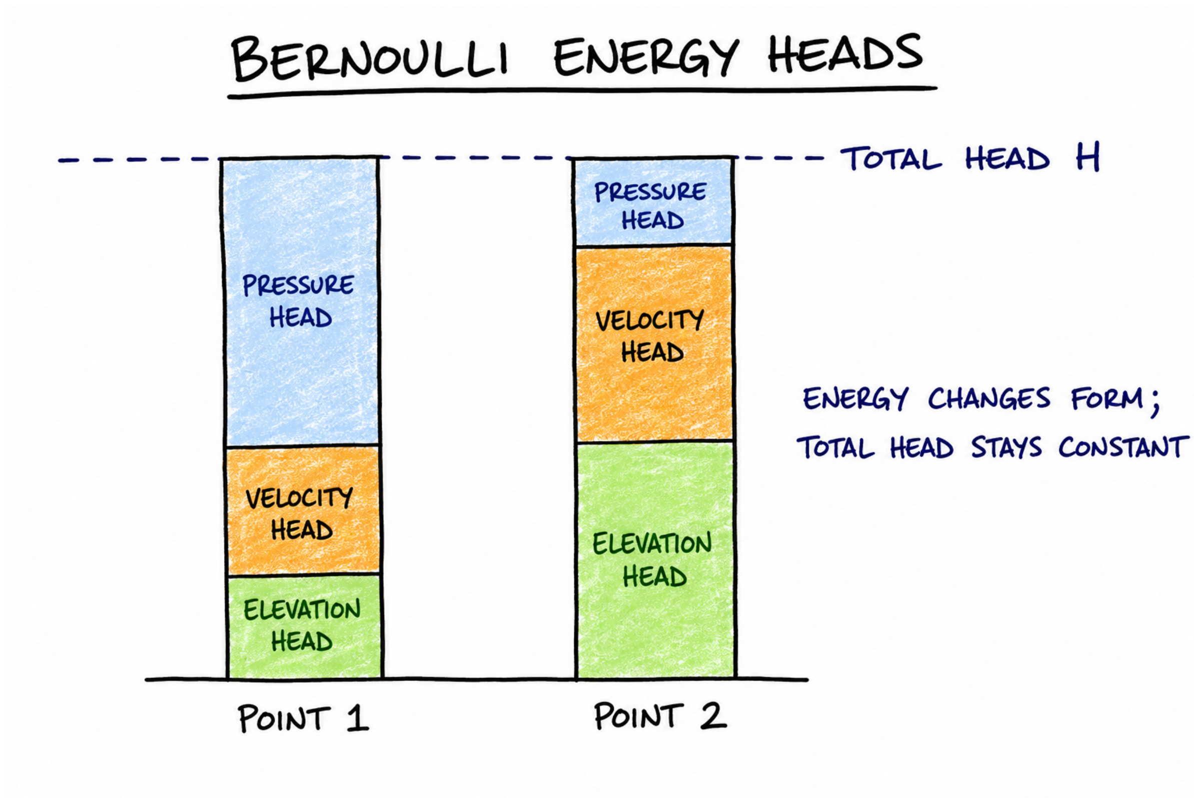
$$\boxed{\frac{P}{\rho} + \frac{V^2}{2} + gz = \text{constant}}$$

Each term has units of $\mathrm{m}^2/\mathrm{s}^2$, equivalent to energy per unit mass.

6 Energy Interpretation

Bernoulli's equation represents conservation of mechanical energy for an ideal fluid particle:

$$\underbrace{\frac{P}{\rho g}}_{\text{pressure head}} + \underbrace{\frac{V^2}{2g}}_{\text{velocity head}} + \underbrace{z}_{\text{elevation head}} = \underbrace{H}_{\text{total head}}.$$



A reduction in pressure can be associated with an increase in velocity, a rise in elevation, or both. The phrase "higher velocity means lower pressure" is only valid when the remaining Bernoulli terms and assumptions are considered.

7 Stagnation Pressure

A stagnation point is a location where the fluid is brought to rest:

$$V_0 = 0.$$

For negligible elevation change, Bernoulli's equation between a moving point and the stagnation point gives

$$P + \frac{1}{2}\rho V^2 = P_0,$$

where P_0 is the stagnation pressure. Thus,

$$\boxed{P_0 - P = \frac{1}{2}\rho V^2}$$

and

$$\boxed{V = \sqrt{\frac{2(P_0 - P)}{\rho}}}$$

This relation is the basis of a Pitot tube.

8 Continuity: Bernoulli's Companion Equation

Bernoulli's equation alone often contains more than one unknown. Conservation of mass supplies another relation:

$$\dot{m} = \rho A V.$$

For steady, incompressible flow through a streamtube,

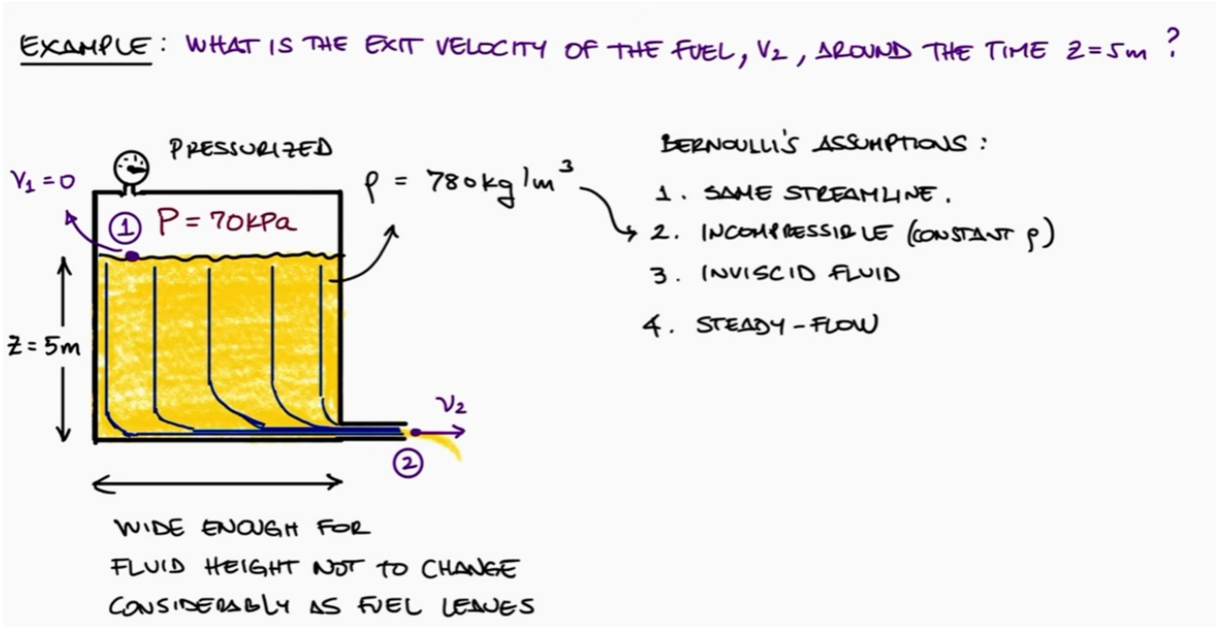
$$\boxed{A_1 V_1 = A_2 V_2}$$

Therefore, when the cross-sectional area decreases, the average velocity increases.

9 Worked Example 1: Pressurized Fuel Reservoir

Problem

A large reservoir contains fuel with density $\rho = 780\,\mathrm{kg/m^3}$. The air above the fuel is maintained at a gauge pressure of $P_1 = 70\,\mathrm{kPa}$. The fuel free surface is $5\,\mathrm{m}$ above a small outlet. Determine the ideal exit velocity.



Choose point 1 at the free surface and point 2 at the outlet. Apply Bernoulli:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2.$$

For a large reservoir, the free-surface area is much larger than the outlet area, so

$$V_1 \approx 0.$$

Use gauge pressure:

$$P_1 = 70\,\text{kPa}, \qquad P_2 = 0.$$

Choose the outlet as the datum:

$$z_1 = 5\,\text{m}, \qquad z_2 = 0.$$

Then

$$V_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho} + 2g(z_1 - z_2)}.$$

Substitution gives

$$V_2 = \sqrt{\frac{2(70000\,\text{Pa})}{780\,\text{kg/m}^3} + 2(9.81\,\text{m/s}^2)(5\,\text{m})}.$$

$$\boxed{V_2 = 16.66\,\text{m/s}}$$

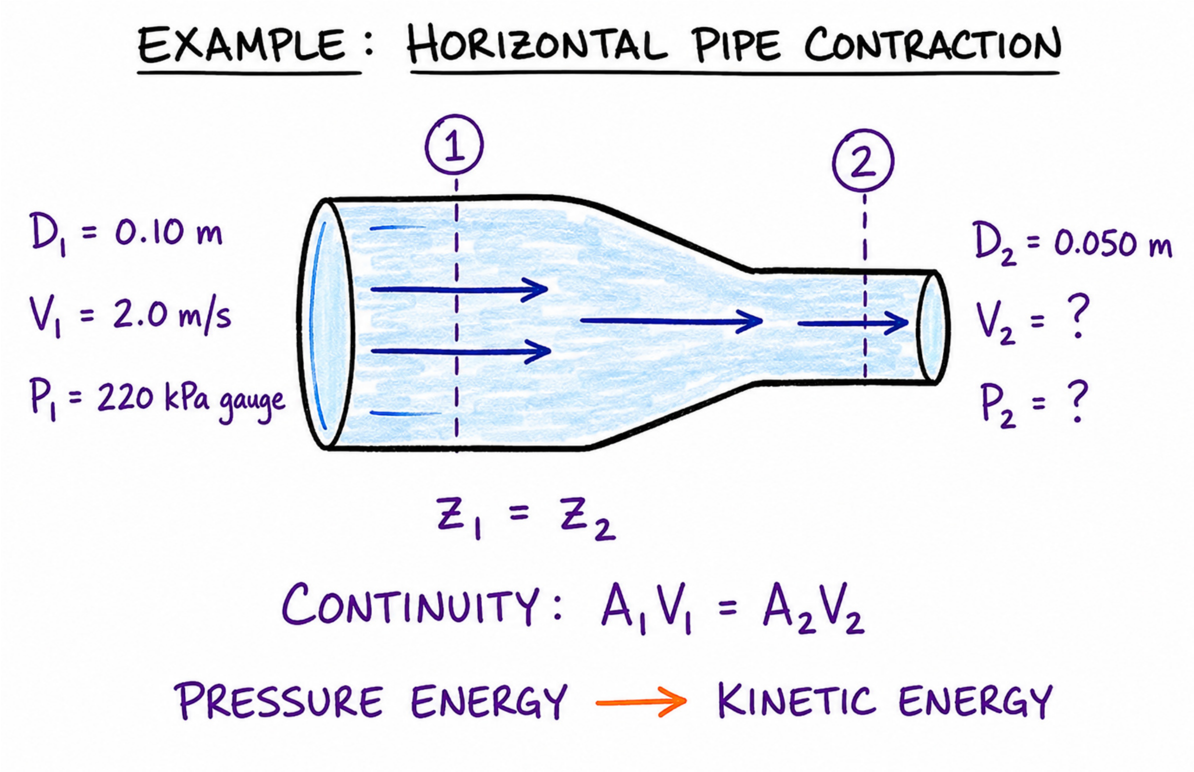
Important Observation

The ideal exit speed depends on the pressure difference and the elevation difference between points 1 and 2. A hose may curve upward and downward between the tank and outlet; if losses are neglected and the outlet elevation is unchanged, the predicted outlet speed is unchanged.

10 Worked Example 2: Horizontal Pipe Contraction

Problem

Water flows steadily through a horizontal reducer. At section 1, $D_1 = 0.10\,\text{m}$, $V_1 = 2.0\,\text{m/s}$, and $P_1 = 220\,\text{kPa}$ gauge. At section 2, $D_2 = 0.050\,\text{m}$. Neglect losses and determine V_2 and P_2.



Continuity gives

$$A_1 V_1 = A_2 V_2,$$

so

$$V_2 = V_1 \frac{A_1}{A_2} = V_1 \left(\frac{D_1}{D_2}\right)^2.$$

$$V_2 = 2.0\,\mathrm{m/s} \left(\frac{0.10}{0.050}\right)^2 = \boxed{8.0\,\mathrm{m/s}}.$$

Because the pipe is horizontal, $z_1 = z_2$. Bernoulli becomes

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2.$$

For water, use $\rho = 1000\,\mathrm{kg/m^3}$:

$$P_2 = P_1 + \frac{1}{2}\rho(V_1^2 - V_2^2).$$

$$P_2 = 220\,\mathrm{kPa} + \frac{1}{2}(1000\,\mathrm{kg/m^3})\left[(2\,\mathrm{m/s})^2 - (8\,\mathrm{m/s})^2\right].$$

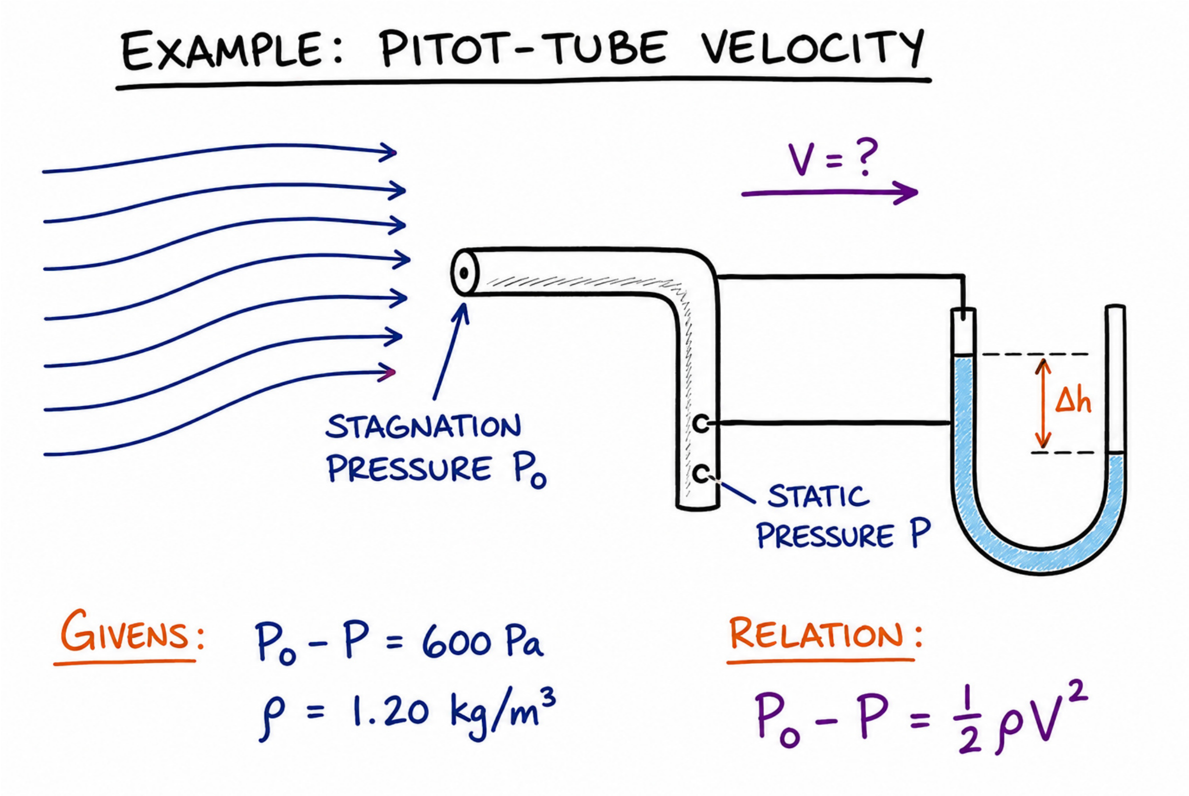
$$\boxed{P_2 = 190\,\mathrm{kPa\ gauge}}$$

The contraction converts part of the pressure energy into kinetic energy.

11 Worked Example 3: Pitot-Tube Velocity

Problem

A Pitot tube in an air stream measures a stagnation pressure that is 600 Pa greater than the static pressure. Take the air density as $\rho = 1.20\,\mathrm{kg/m^3}$. Determine the air speed.



For negligible elevation difference,

$$P_0 - P = \frac{1}{2}\rho V^2.$$

Thus,

$$V = \sqrt{\frac{2(P_0 - P)}{\rho}} = \sqrt{\frac{2(600\,\mathrm{Pa})}{1.20\,\mathrm{kg/m^3}}}.$$

$$\boxed{V = 31.6\,\mathrm{m/s}}$$

12 A Reliable Problem-Solving Procedure

1. Draw a clear control-volume or streamline sketch.
2. Select points 1 and 2 where the most information is known.
3. State the Bernoulli assumptions explicitly.
4. Choose a convenient elevation datum.
5. Decide whether pressure values are absolute or gauge; do not mix them.

6. Write the full Bernoulli equation before canceling terms.
7. Use continuity if velocity and area are related.
8. Substitute values with units.
9. Check whether the result is physically reasonable.

Common Simplifications

Situation	Simplification
Large reservoir free surface	$V_1 \approx 0$
Free jet exposed to atmosphere	$P = 0$ gauge
Horizontal pipe	$z_1 = z_2$
Same pipe diameter	$V_1 = V_2$ for incompressible flow
Stagnation point	$V = 0$
Both points open to atmosphere	$P_1 = P_2$

13 Common Mistakes

1. **Mixing gauge and absolute pressure.** Use one pressure convention consistently.
2. **Canceling pressure terms when only one point is atmospheric.**
3. **Assuming $V = 0$ everywhere inside a tank.** Only the large free-surface velocity is often negligible.
4. **Using diameter instead of area in continuity.** Since $A \propto D^2$, the velocity ratio depends on the square of the diameter ratio.
5. **Ignoring head loss in a long or rough pipe.**
6. **Using two points that are not connected by a valid streamline in rotational flow.**
7. **Saying that pressure always decreases when velocity increases.** Elevation and energy additions or losses also matter.

14 In-Class Concept Questions

Check Your Understanding

1. Water exits a large open tank through a small hole. Which terms in Bernoulli's equation cancel or become negligible?
2. A horizontal pipe narrows. For ideal incompressible flow, what happens to velocity and static pressure?
3. A Pitot tube measures $P_0 - P$. What physical quantity does this pressure difference represent?
4. Can the basic Bernoulli equation be written directly across a pump? Explain.
5. In the reservoir example, why is gauge pressure convenient?

15 Practice Problems

Problem 1: Open Tank

Water exits from a small opening located 3.0 m below the free surface of a large open tank. Neglect losses. Determine the exit speed.

Problem 2: Nozzle

Water enters a horizontal nozzle at $D_1 = 80\,\text{mm}$, $V_1 = 2.5\,\text{m/s}$, and $P_1 = 180\,\text{kPa}$ gauge. The outlet diameter is $D_2 = 40\,\text{mm}$. Determine V_2 and the ideal outlet pressure.

Problem 3: Vertical Pipe

Water flows upward through a constant-diameter pipe. Point 2 is 6 m above point 1. If $P_1 = 250\,\text{kPa}$ gauge and losses are negligible, determine P_2.

Problem 4: Pitot Tube

A Pitot tube in water measures $P_0 - P = 8.0\,\text{kPa}$. Determine the local water velocity.

Problem 5: Pressurized Tank

A large tank contains oil with density $850\,\text{kg/m}^3$. The gas pressure above the oil is 45 kPa gauge. An outlet is 4 m below the free surface. Determine the ideal exit speed.

16 Answers to Practice Problems

1.
$$V = \sqrt{2gh} = \sqrt{2(9.81\,\text{m/s}^2)(3.0\,\text{m})} = \boxed{7.67\,\text{m/s}}.$$

2.

$$V_2 = V_1\left(\frac{D_1}{D_2}\right)^2 = 2.5\,\mathrm{m/s}(2)^2 = \boxed{10.0\,\mathrm{m/s}}.$$

$$P_2 = P_1 + \frac{1}{2}\rho(V_1^2 - V_2^2) = \boxed{133.1\,\mathrm{kPa\ gauge}}.$$

3. Since the diameter is constant, $V_1 = V_2$:

$$P_2 = P_1 - \rho g(z_2 - z_1) = \boxed{191.1\,\mathrm{kPa\ gauge}}.$$

4.

$$V = \sqrt{\frac{2(P_0 - P)}{\rho}} = \sqrt{\frac{2(8000\,\mathrm{Pa})}{1000\,\mathrm{kg/m^3}}} = \boxed{4.00\,\mathrm{m/s}}.$$

5.

$$V = \sqrt{\frac{2(45000\,\mathrm{Pa})}{850\,\mathrm{kg/m^3}} + 2(9.81\,\mathrm{m/s^2})(4\,\mathrm{m})} = \boxed{13.58\,\mathrm{m/s}}.$$

17 Summary

Key Takeaways

- Along a streamline, pressure, kinetic, and potential energy can exchange with one another.
- For steady, incompressible, inviscid flow with no shaft work,

$$P + \frac{1}{2}\rho V^2 + \rho g z = \text{constant}.$$

- The head form is

$$\frac{P}{\rho g} + \frac{V^2}{2g} + z = \text{constant}.$$

- Continuity, $A_1V_1 = A_2V_2$, is often needed together with Bernoulli.
- Assumptions must be checked before using the equation.
- Real pipe-flow problems often require head-loss and pump/turbine terms.